

CyberShield: A 2-Tier Cascade Network Intrusion Detection System (NIDS)

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Abstract—CyberShield is a 2-Tier Cascade Network Intrusion Detection System (NIDS) designed to address extreme class imbalance, dataset leakage, and real-world deployment constraints in cybersecurity systems. Built on the CICIDS2017 dataset, the system undergoes a rigorous data-engineering pipeline that removes duplicate records, eliminates data leakage, and fixes label inconsistencies, resulting in a clean dataset of 2,520,798 network flows.

The architecture consists of Tier B (Random Forest) for fast multi-class classification and Tier C (XGBoost) for deep rare-attack detection. The system achieves high Macro-F1 performance while maintaining real-time inference capability.

Deployment is hybrid: a C++ ONNX Runtime engine runs on Android for sub-2ms edge inference, while a FastAPI service on AWS Elastic Beanstalk handles cloud requests.

Index Terms—Intrusion Detection System, CICIDS2017, Random Forest, XGBoost, Class Imbalance, ONNX Runtime, Edge AI, Cybersecurity

I. INTRODUCTION

Modern networks generate massive traffic, requiring intelligent intrusion detection systems. Signature-based IDS fail against unknown attacks, motivating machine learning-based approaches.

However, datasets like CICIDS2017 suffer from severe class imbalance, where benign traffic dominates rare attack types such as Heartbleed and Infiltration.

CyberShield addresses this using a 2-Tier Cascade architecture optimized for accuracy, efficiency, and deployment feasibility.

Key Contributions:

- Cleaned CICIDS2017 dataset with leakage removal
- 2-Tier Cascade ML architecture (RF + XGBoost)
- Hybrid Edge/Cloud deployment system

II. RELATED WORK

Random Forest models are highly effective for imbalanced tabular data due to ensemble robustness [1]. SMOTE introduces synthetic bias in high-dimensional datasets [2]. ONNX Runtime enables efficient cross-platform deployment [3].

III. DATASET AND DATA ENGINEERING

CICIDS2017 contains 2.83 million flow records with 78 network features.

A. Issues Identified

- Data leakage via Is_Attack feature
- 307,078 duplicate records
- Label inconsistencies across models

B. Final Dataset Distribution

TABLE I
CLEAN CICIDS2017 DATASET

Attack Type	Count	%
BENIGN	2,018,340	80.08
DoS Hulk	205,139	8.14
PortScan	141,259	5.60
DDoS	113,705	4.51
Others	Remaining	12

Total records: **2,520,798**

IV. SYSTEM ARCHITECTURE

A. Tier B: Random Forest

- 75 estimators
- max depth 25
- balanced class weighting
- fast general attack detection

B. Tier C: XGBoost

- 200 estimators
- learning rate 0.1
- histogram-based training
- sample-weighted rare attack detection

C. Cascade Logic

- 1) Tier B predicts initial class
- 2) If confidence $\geq 0.70 \rightarrow$ escalate to Tier C
- 3) Tier C produces final decision

V. DEPLOYMENT ARCHITECTURE

A. Edge Deployment (Android)

- C++ ONNX Runtime engine
- Direct memory inference
- Sub-2ms latency
- Fully offline execution

B. Cloud Deployment (AWS)

- FastAPI microservice
- REST API endpoint (/classify)
- Scalable cloud inference

VI. EXPERIMENTAL RESULTS

TABLE II
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Macro F1	Heartbleed	Infiltration
MLP (Leaky)	99.99*	0.0669	0.00	0.00
MLP (Clean)	82.41	0.1134	0.00	0.00
Tier B RF	99.83	0.8605	0.50	1.00
Tier C XGB	99.86	0.8821	1.00	1.00

**Accuracy inflated due to data leakage*

VII. CONCLUSION

CyberShield demonstrates a robust 2-Tier Cascade NIDS using Random Forest and XGBoost. The system achieves high accuracy, strong macro-F1 performance, and real-world deployment capability across edge and cloud environments.

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